

When Users (Pleasantly) Surprise Us: The Joy of Unveiling Hidden Insights through User Studies with Specific Participant Groups

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User studies with specific participant groups, such as children, people with developmental disabilities, and domain professionals, can reveal discrepancies between researchers' expectations and participants' behavior. Understanding these discrepancies in light of participants' contexts can lead to new findings and opportunities to improve systems and studies. *When Users (Pleasantly) Surprise Us* brings together researchers and practitioners to share unexpected participant behavior, how they responded, and what they learned. Using case cards, attendees will exchange experiences in two rounds of small-group discussion, first with peers studying similar participant groups and then with those studying different groups. The meet-up aims to help researchers learn from one another's experiences and turn unexpected participant behavior into opportunities for discovery and improvement in systems and studies involving specific participant groups.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**; **User studies**.

Additional Key Words and Phrases: User Studies, Participant Behavior, Specific Participant Groups, Researcher Assumptions

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1 Motivation and Goals

Researchers design user studies with expectations about how participants will interpret tasks and interact with a system. However, participants often behave in unexpected ways, such as finding shortcuts that bypass intended steps or using a feature for an unintended purpose. These discrepancies can produce unexpected results or even study failure, but can also lead to new findings and opportunities to improve the system and its evaluation.

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Table 1. An Example of participant-group categories for badges, adapted from CHI's user descriptors. Categories may overlap.

Group category	Examples of participant groups
Age and life stage	Children and parents, teens, older adults
Disability and accessibility	People with developmental, sensory, or physical disabilities
Professional and social roles	Clinicians, therapists, designers, athletes, students, and educators
Sociocultural and resource contexts	Racialized or culturally minoritized communities, and communities with limited resources

Such discrepancies can be particularly challenging to understand in user studies with specific participant groups, such as people with developmental disabilities, children, or domain-specific professionals. Limited familiarity with these groups' communication styles, priorities, or everyday practices can make it difficult for researchers to fully anticipate how participants will engage with a study. Interpreting unexpected behavior in light of participants' contexts can therefore help researchers revisit their assumptions and identify new insights and opportunities for improvement.

When Users (Pleasantly) Surprise Us meet-up will provide a space for researchers to exchange how they interpreted unexpected participant behavior, responded to it, and learned from these experiences. Through this exchange, attendees can get practical insights and perspectives to inform their own systems and studies. Attendees will first discuss cases with peers studying similar participant groups, then compare perspectives across groups and contexts. We welcome difficult experiences, helpful adjustments, positive surprises, and unresolved questions. Case cards and badges indicating research experience or interests will support conversations and connections, including for people preparing their first user study.

The meet-up aims to help researchers learn from unexpected participant behavior while supporting continued exchange across research communities. Specifically, our goals are to (1) share concrete cases and how researchers interpreted and responded to unexpected behavior in light of participants' contexts; (2) draw on experiences across participant groups to reconsider assumptions and recognize potential insights and opportunities for improving systems and studies; and (3) connect people with related questions and complementary experiences so that conversations can continue beyond the session.

2 Planned Activities

The 90-minute, in-person session will use *case cards* and badges to connect two rounds of small-group discussion. It is free and open to all registered CHI attendees, with no advance preparation or submission. Case cards are optional. Organizers will provide materials and orient late arrivals before helping them join a group.

Welcome and badges (0–10 minutes). On arrival, attendees will receive a badge and lanyard and name participant groups they have studied, using the categories adapted from CHI's user descriptors in Table 1 as a guide. Those who have studied only general populations or have no study experience can indicate a group of interest. Organizers will introduce the goals and an example, inviting discussion of participants' perspectives without identifying details.

Preparing a case card (10–20 minutes). Attendees may note their system or task and participant context, then use the three Activity 1 questions to outline a case. An unresolved situation or question is welcome. Cards will support introductions, discussion, and group formation.

Activity 1: Stories within Participant Contexts (20–45 minutes). Attendees will form groups of four to six around similar participant groups or study contexts. Those without relevant experience may join a group of interest. Cases will begin with two-minute introductions, followed by questions and discussion: (1) *What did you expect, and*

what did the participant do? (2) How did you respond, and what informed your understanding? (3) What did you learn, or what remains unresolved? This round will help attendees understand one another's situations and exchange practical knowledge grounded in related participant contexts.

Regrouping (45–50 minutes). Organizers will use badges and cards to mix experiences across participant groups, balancing group sizes and people able to share cases. Newcomers may follow their interests. If experiences overlap, groups will compare different tasks or settings. Attendees may remain seated while others move.

Activity 2: Perspectives across Participant Groups (50–75 minutes). Attendees will share the same case cards with their new groups, followed by questions and informal discussion. This round will offer opportunities to hear unfamiliar study experiences and discover interesting similarities and differences in the unexpected situations researchers encountered, how they responded, and what they learned. Attendees will also leave comments, questions, or suggestions in the designated space on one another's case cards. These notes can offer insights for future user studies and starting points for further conversation.

Sharing reflections (75–90 minutes). Each group will briefly share one assumption or question that emerged. Organizers will connect related contributions and reserve time for attendees to continue conversations and exchange contact details if they wish.

3 Organizers

The nine organizers of this meet-up have expertise in robotics and HCI and extensive experience conducting user studies with diverse participant groups. **Minwoo Oh** is interested in VR/AR applications in dynamic settings, including vehicles and sports [3, 7], while conducting a user study collecting golf swing sensor data from professional golfers. **Chaeyeon Lim** has researched devices supporting social interaction among children with developmental disabilities and conducted user studies with these children [1]. **Minwoo Seong** has conducted semi-structured interviews and user studies collecting sensor data for badminton swing analysis with badminton experts [5]. **Jeongseok Oh** has conducted user studies of generative AI systems with game development and 3D modeling professionals. **Kangbeen Ko** has conducted user studies with therapists specializing in children with developmental disabilities. **SeungJun Kim** is a professor at GIST, and his research has included user studies evaluating participants' VR motion sickness and presence with visual, auditory, and haptic feedback in moving vehicles. **Won Kim** is interested in HCI in physical AI and XR environments and researches interactive systems supporting the engagement of children with autism spectrum disorder [1, 2]. **Jin Ryong Kim** is an associate professor in the Department of Computer Science at the University of Texas at Dallas. He is interested in haptics and multimodal interaction in VR and researches interfaces using thermal and tactile feedback [6]. **Daniela Rus** is a professor of computer science at the Massachusetts Institute of Technology (MIT) and director of the Computer Science and Artificial Intelligence Laboratory (CSAIL). She is interested in robotics and human-robot collaboration and researches robot interfaces that reflect human intent [4].

4 Who We Hope to Bring Together

We especially welcome researchers, practitioners, and students who have conducted user studies with specific participant groups, including groups defined by age or life stage, disability, professional or social roles, and sociocultural or resource contexts. We also welcome people who have helped run study sessions or worked closely with these participant groups, as well as researchers preparing to work with a group that is new to them. People who have studied general populations or have no user study experience are equally welcome to share experiences, ask questions, listen, and connect with peers.

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